Letter for Number

Year 4 · **Lesson Worksheet**

Worked Examples

Example 1: Using A=1, B=2, C=3 … Z=26, what is the value of F + K?

Answer: 17 — F = 6 and K = 11, so 6 + 11 = 17.

Example 2: Using A=1, B=2, … Z=26, decode the word given by the numbers 4-15-7.

Answer: DOG — 4 = D, 15 = O, 7 = G.

Example 3: If A=5, B=6, C=7 … (each letter's value is 4 more than its usual alphabet position), what is the value of the letter D?

Answer: 8 — D is normally the 4th letter, and 4 + 4 = 8.

Purple — Accessible (15 questions)

1. Using A=1, B=2, C=3 … Z=26, what is the value of the letter J?
2. Using A=1, B=2, … Z=26, what letter has the value 14?
3. Using A=1, B=2, … Z=26, what is the value of the letter M? A) 11 B) 12 C) 13 D) 14 E) 15
4. Using A=1, B=2, … Z=26, what is B + E?
5. Using A=1, B=2, … Z=26, what is H − C?
6. Using A=1, B=2, … Z=26, decode the word given by the numbers 3-1-20. A) BAT B) CAT C) CAR D) BAR E) CAN
7. Using A=1, B=2, … Z=26, encode the word DOG as a sequence of numbers.
8. Using A=1, B=2, … Z=26, what is the value of the letter R?
9. Using A=1, B=2, … Z=26, select ALL the letters below with a value GREATER than 20. A) S B) U C) W D) T E) V
10. Using A=1, B=2, … Z=26, decode the word given by the numbers 19-21-14.
11. Using A=1, B=2, … Z=26, what is F + G? A) 11 B) 12 C) 13 D) 14 E) 15
12. Using A=1, B=2, … Z=26, what is the value of the letter Y?
13. Using A=1, B=2, … Z=26, encode the word CAT as a sequence of numbers.
14. Using A=1, B=2, … Z=26, what is P − J? A) 4 B) 5 C) 6 D) 7 E) 8
15. Using A=1, B=2, … Z=26, decode the word given by the numbers 2-9-7.

Green — Standard (15 questions)

1. Using A=1, B=2, … Z=26, what is C + H + M?
2. Using A=1, B=2, … Z=26, what is the value of Z − A?
3. Using A=1, B=2, … Z=26, decode the word given by the numbers 16-12-1-14. A) PLAT B) PLAN C) PLEN D) PLON E) PLIN

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4. Using A=1, B=2, ... Z=26, what is the SUM of the values of the letters in the word CAB?

5. Using A=1, B=2, ... Z=26, encode the word STAR as a sequence of numbers.

6. Using A=1, B=2, ... Z=26, what is K × 2? A) 20 B) 22 C) 24 D) 26 E) 28

7. Using A=1, B=2, ... Z=26, what is the value of the letter that comes immediately after N in the alphabet?

8. Using A=1, B=2, ... Z=26, what is the DIFFERENCE between the values of W and Q?

9. Using A=1, B=2, ... Z=26, select ALL the words below whose letters have a total SUM greater than 30. A) CAT (3,1,20) B) DOG (4,15,7) C) BAT (2,1,20) D) SUN (19,21,14) E) PIG (16,9,7)

10. Using A=1, B=2, ... Z=26, what is the value of the MIDDLE letter of the word FOX?

11. Using A=1, B=2, ... Z=26, what is the value of V + V? A) 40 B) 42 C) 44 D) 46 E) 48

12. Using A=1, B=2, ... Z=26, decode the word given by the numbers 7-18-5-5-14.

13. Using A=1, B=2, ... Z=26, what is the value of the letter that comes 3 places AFTER J in the alphabet?

14. Using A=1, B=2, ... Z=26, what is the SUM of the values of the letters in the word TEN? A) 37 B) 38 C) 39 D) 40 E) 41

15. Using A=1, B=2, ... Z=26, encode the word MOON as a sequence of numbers.

Red — Challenge (15 questions)

1. A code gives every letter a value 4 higher than its spot in the alphabet (so A=5, B=6, C=7 and so on). What value does this code give the letter E?

2. Keep that same +4 code from the last question. What value does it give the letter K?

3. A different code gives each letter a value 2 higher than its spot in the alphabet (A=3, B=4, C=5...). What value does F get? A) 7 B) 8 C) 9 D) 10 E) 11

4. Under a code where every letter is worth 1 LESS than its position (A=0, B=1, C=2...), what is P worth?

5. Still using that "1 less" code (A=0, B=1, C=2...), which letter is worth 10?

6. Flip the alphabet back to front, so Z=1, Y=2, X=3 and so on (making A=26). On this scale, what is D worth? A) 21 B) 22 C) 23 D) 24 E) 25

7. On that same flipped scale (Z=1 ... A=26), which letter is worth 5?

8. Go back to the +4 code (A=5, B=6, C=7...). Add together the values this code gives to F and G.

9. A code gives every letter a value 11 higher than its spot in the alphabet (A=12, B=13...). Select ALL the letters below worth 20 or MORE on this code. A) H B) J C) K D) D E) F

10. On that same +11 code (A=12, B=13...), what is Z worth?

11. A code gives each letter DOUBLE its usual alphabet position (A=2, B=4, C=6...). What does this code give J? A) 18 B) 20 C) 22 D) 24 E) 26

12. On that same doubling code (A=2, B=4, C=6...), which letter is worth 16?

13. Standard code A=1, B=2 ... Z=26 turns FOX into 6-15-24. Add 1 to each of those three numbers. What word do the new numbers spell?

14. Standard code A=1, B=2 ... Z=26 turns LAMP into 12-1-13-16. Add 1 to each of those four numbers. What word do the new numbers spell? A) MBNQ B) MBNR C) LBNQ D) MANQ E) MBMQ

15. Standard code A=1, B=2 ... Z=26. Find a single letter that, added to D's value, gives exactly J's value.

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